人工智能导论

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个人简介

- 教育经历
  - 本科 计算机科学与技术系 南京大学 2011-2015
  - 硕士 计算机科学与技术系 南京大学 2015-2018
  - 博士 计算机科学与技术系 南京大学 2018-2022
- 工作经历
  - 助理研究员/副研究员 计算机学院 2022.11-2026.5
  - 助理教授（特聘研究员/博导）智能科学与技术学院 2026.6-至今
- 教学经历
  - 计算机系统基础理论课
  - 计算机系统基础实验课（PA实验）AI/计算机/医院

Jie Liu (刘杰)

助理教授 (特聘研究员) · 博导
School of Intelligence Science and Technology
Nanjing University, Suzhou · Jiangsu, China

> 🐳 I am recruiting **1–2 Ph.D. students and 1–2 master's students for 2027**. Interested students are welcome to contact me at liujie@nju.edu.cn.

https://njulj.github.io/

Biography

I am working as an Assistant Professor at the School of Intelligence Science and Technology, Nanjing University. I am also a member of the Multimedia Computing Group (MCG), working with Prof.Jie Tang and Prof.Limin Wang. Previously, I obtained my Ph.D. at the Department of Computer Science and Technology, Nanjing University, China, in Sep. 2022. Before that, I received my master's degree in the Department of Computer Science and Technology at Nanjing University, China, in Jue. 2018 and B.E degree from the Department of Computer Science and Technology, Nanjing University, China, in Jue. 2015.

My research interest broadly includes deep learning, computer vision and speech processing. Specifically, I focus on image/video restoration and generation, multimodal perception and understanding, efficient MLLM/LLM, agentic system, etc.

Current Research

Controllable Image Generation (AIGC)
Generative Image Restoration/Super-Resolution/Editing

Spatiotemporal Intelligence (AI4Sci)
Climate downscaling and AI for meteorology

Understanding Complex Scenes (MLLM)
MLLM Learning in degraded or complex scenarios

Audio-Visual Multimodal Interaction
Interaction and Navigation in Embodied Scenarios

Agentic Systems
Agents addressing all the research topics above

考核方式

- 期末考试 45%
- 课程实践 45%
    - Assignment 1　10%
    - Assignment 2　10%
    - Assignment 3　10%
    - Final Project　15%
- 课程竞赛 10%
    - Competition　（提交获得分数即可拿到80%成绩，另外20%为报告分数， private leaderboard前30可加分）
- DDL规则，每次实践会发布soft和hard ddl
    - Soft ddl前提交奖励5%，hard ddl前提交无奖励不扣分，hard ddl后提交只算80%分数
    - **实践+竞赛部分汇总后总分数不超过55**
- 所有实践/竞赛作业成绩会通过NJU Table给每位同学反馈，注意查收

关于大模型的使用

- 实践作业不限制使用大模型，但是要提供使用大模型的对话记录
- 学习知识不是最重要的，知识可以通过大模型获取
- 重要的是理解创造知识的人如何思考和取舍（Trade-off）

课程群

人工智能导论26

群号: 1106622528

扫一扫二维码，加入群聊

QQ

课程主页

- https://njulj.github.io/courses/
- 课件只发布在课程群中，重要消息会在课程主页和课程群中同步发布

助教

- 李志梁（博士）652024330015@smail.nju.edu.cn
- 谢鹏宇（硕士）522026330103@smail.nju.edu.cn
- 雷业成 （大三）241220106@smail.nju.edu.cn
- 徐子轩（大三）241220035@smail.nju.edu.cn

参考学习资料

- https://www.bilibili.com/video/BV1TAtwzTE1S/?spm_id_from=333.337.search-card.all.click

Competition

- RSNA Knee Abnormality Detection AI Challenge (2026)
- 官网： https://www.rsna.org/artificial-intelligence/ai-image-challenge/knee-mri-ai-challenge?utm_source=chatgpt.com
- Kaggle比赛链接： https://www.kaggle.com/competitions/rsna-knee-abnormality-detection/overview

Prizes

Main Leaderboard
First Prize: $9,000
Second Prize: $7,000
Third Prize: $6,500
Fourth Prize: $6,000
Fifth Prize: $5,500
Sixth Prize: $5,000
Seventh Prize: $5,000
Eighth Prize: $5,000
Ninth Prize: $5,000
Tenth Prize: $5,000

Efficiency Track
First Efficiency Prize: $7,000
Second Efficiency Prize: $6,000
Third Efficiency Prize: $5,000

课程竞赛 10%

Competition （提交获得分数即可获得70%成绩，另外30%为报告分数，Main/Efficiency track获奖可加5分，Private Leaderboard排名11-20加2分，21-30加1分）

Team name请使用自己的学号

How Can AI Help Evaluate Knee Injuries?

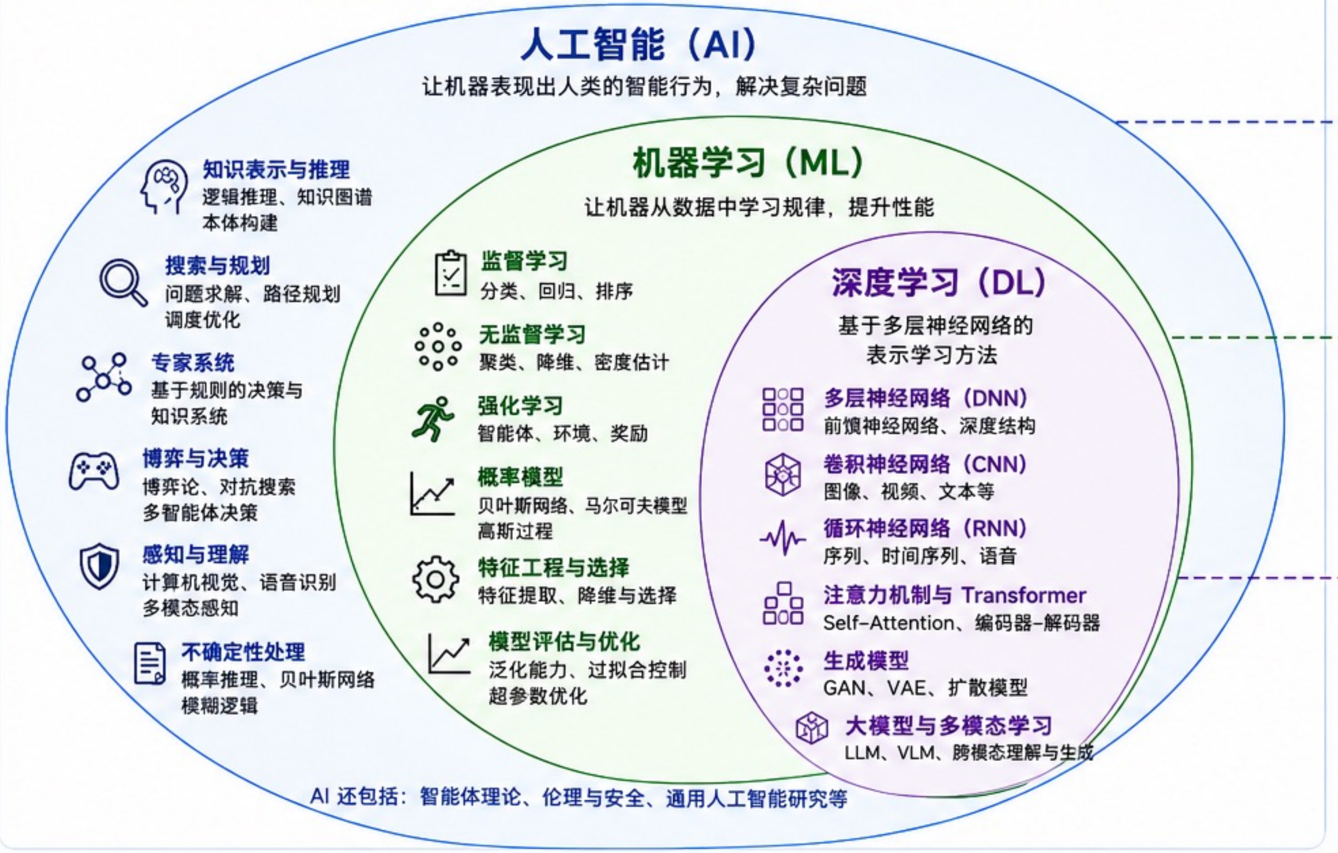
While MRI reveals damage to joint structures, including ligaments, menisci and cartilage loss, interpretation varies and access to specialty-trained musculoskeletal radiologists is limited. By training AI models on a large dataset of knee MRIs paired with their original multilingual reports, this challenge aims to develop tools that can reliably identify abnormalities and support more consistent, timely care.

Challenge Timeline and Recognition for Top Performing Teams

The competition is open now and runs until October. The top performing teams will share a total of $77,000 in prize money. Winners will be announced in November and the top performing teams will be recognized during RSNA 2026 (Nov. 29 – Dec. 3) at McCormick Place in Chicago.

Join the Competition →

人工智能、机器学习与深度学习的关系



人工智能发展时间线

1950s – 1970s 人工智能兴起
- 1956：达特茅斯会议，AI 概念提出
- 1960s：符号主义、逻辑推理、搜索算法
- 1970s：专家系统兴起，早期 AI 应用

1980s – 2000s 机器学习发展
- 1980s：统计学习理论发展，贝叶斯方法
- 1990s：SVM、决策树、集成学习等算法
- 2000s：大规模数据、核方法、图模型

2010s 深度学习突破
- 2012：AlexNet 在 ImageNet 大赛夺冠
- 2014：Seq2Seq、注意力机制提出
- 2015–2016：ResNet、GAN 等突破
- 2017：Transformer 架构提出

2020s – 至今 基础模型与生成式 AI
- 2020：大规模预训练模型（GPT-3）
- 2021–2022：多模态模型（CLIP、DALL·E）
- 2023–2024：GPT-4/5、扩散模型、AI Agent
- 通向通用人工智能（AGI）的长期目标

代表性方法（示例）

人工智能（非 ML 方法）	机器学习（经典方法）	深度学习（典型模型）
A* 搜索算法	线性回归 / 逻辑回归	CNN / ResNet（视觉）
Mininmax 博弈搜索	决策树 / 逻辑回归	RNN / LSTM / GRU（序列）
逻辑推理（谓词逻辑）	支持向量机（SVM）	Attention Mechanism
规则系统（IF-THEN 规则）	k-近邻（k-NN）	Transformer
知识图谱构建	k-means 聚类	BERT / GPT（预训练模型）
专家系统 / 基于规则推理	朴素贝叶斯 / 贝叶斯网络	GAN / Diffusion（生成模型）

主要研究内容

人工智能（AI）	机器学习（ML）	深度学习（DL）
• 知识表示与推理	• 监督学习（分类、回归）	• 深度表示学习
• 搜索与规划	• 无监督学习（聚类、降维）	• 卷积网络（视觉）
• 专家系统与规则学习	• 强化学习（策略学习）	• 循环网络（序列）
• 感知与理解	• 概率模型与统计学习	• 注意力机制与 Transformer
• 不确定性处理	• 特征工程与选择	• 生成模型与扩散模型
• 博弈与决策	• 模型评估与优化	• 大模型与多模态学习

总体趋势：从 规则驱动 → 统计学习 → 表示学习 → 大模型与通用智能

数据规模 ↑ | 计算能力 ↑ | 模型复杂度 ↑ | 能力边界不断拓展

课程内容

1. | 绪论
2. | 知识表示与知识图谱
3. | 确定性推理方法
4. | 不确定性推理方法
5. | 搜索策略
6. | 智能计算及其应用
7. | 机器学习与专家系统
8. | 人工神经网络与深度学习
9. | 循环神经网络与自然语言理解
10. | 大语言模型及其应用
11. | AI智能体与多智能体系统
12. | 人工智能在游戏设计中的应用

附录：附录A 部分习题解答 | 附录B 实验指导书 | 参考文献

- Ch0 | 人工智能简介
- Ch1 | 机器学习基础
- Ch2 | 深度学习基础
- Ch3 | 卷积神经网络
- Ch4 | 循环神经网络
- Ch5 | 注意力与Transformer
- Ch6 | 自然语言处理中的自监督学习
- Ch7 | 计算机视觉中的自监督学习
- Ch8 | 生成式模型基础
- Ch9 | 生成式模型进阶
- Ch10 | 分词与大语言模型
- Ch11 | 多模态大模型
- Ch12 | 大模型微调
- Ch13 | 具身智能与智能体

课件来源，感谢下面所有课程！

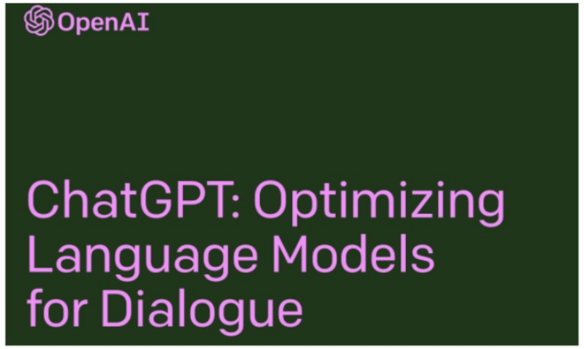
章节	参考课程	学校
Chapter0	CS 15-281: AI—Representation and Problem Solving	Carnegie Mellon University
Chapter1	CS50's Introduction to Artificial Intelligence with Python	Harvard University
Chapter2	6.S191: Introduction to Deep Learning	MIT
Chapter3	6.S191: Introduction to Deep Learning	MIT
Chapter4	6.S191: Introduction to Deep Learning	MIT
Chapter5	CS224N / Ling 284: Natural Language Processing with Deep Learning	Stanford University
Chapter5	CME 295: Transformers & Large Language Models	Stanford University
Chapter6	CS229: Machine Learning	Stanford University
Chapter7	CS231n: Deep Learning for Computer Vision	Stanford University
Chapter8	CS231n: Deep Learning for Computer Vision	Stanford University
Chapter9	CS231n: Deep Learning for Computer Vision	Stanford University
Chapter10	11-785: Introduction to Deep Learning	Carnegie Mellon University

特别致谢

- 以下课件来自CS 15-281: AI—Representation and Problem Solving
- From Carnegie Mellon University

What is AI?

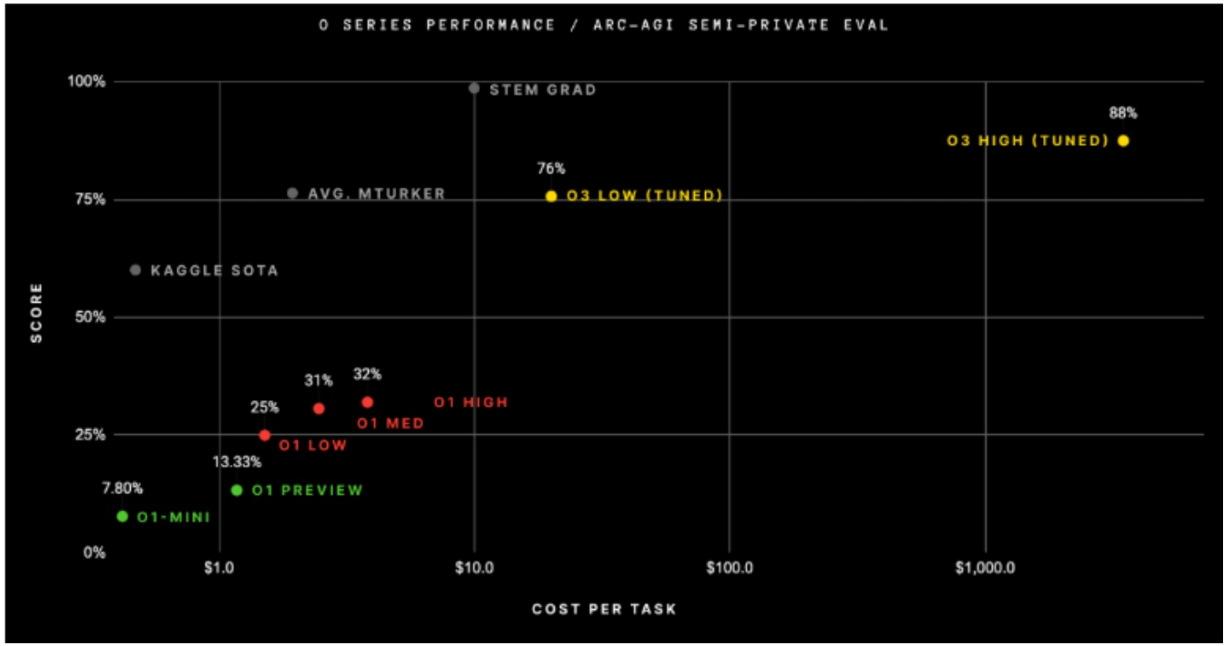
Measurable progress in AI



NEWS | 12 January 2023

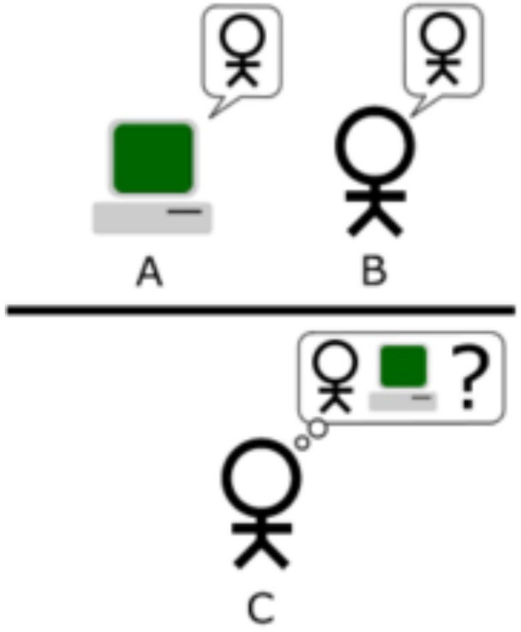
Abstracts written by ChatGPT fool scientists

Researchers cannot always differentiate between AI-generated and original abstracts.



Turing Test

- In 1950, Turing defined a test of whether a machine could "think"
- "A human judge engages in a natural language conversation with one human and one machine, each of which tries to appear human. If judge can't tell, machine passes the Turing test"



en.wikipedia.org/wiki/Turing_test

Some classic definitions

Think like a human

- Cognitive science / neuroscience
- Does intelligence require biology?

Think rationally

- Logic and automated reasoning
- But, not all problems can be solved just by these techniques

Act like a human

- Turing test
- ELISA, Loebner prize
- "What is 1228 x 5873"? … "I don't know, I'm just a human

Act rationally

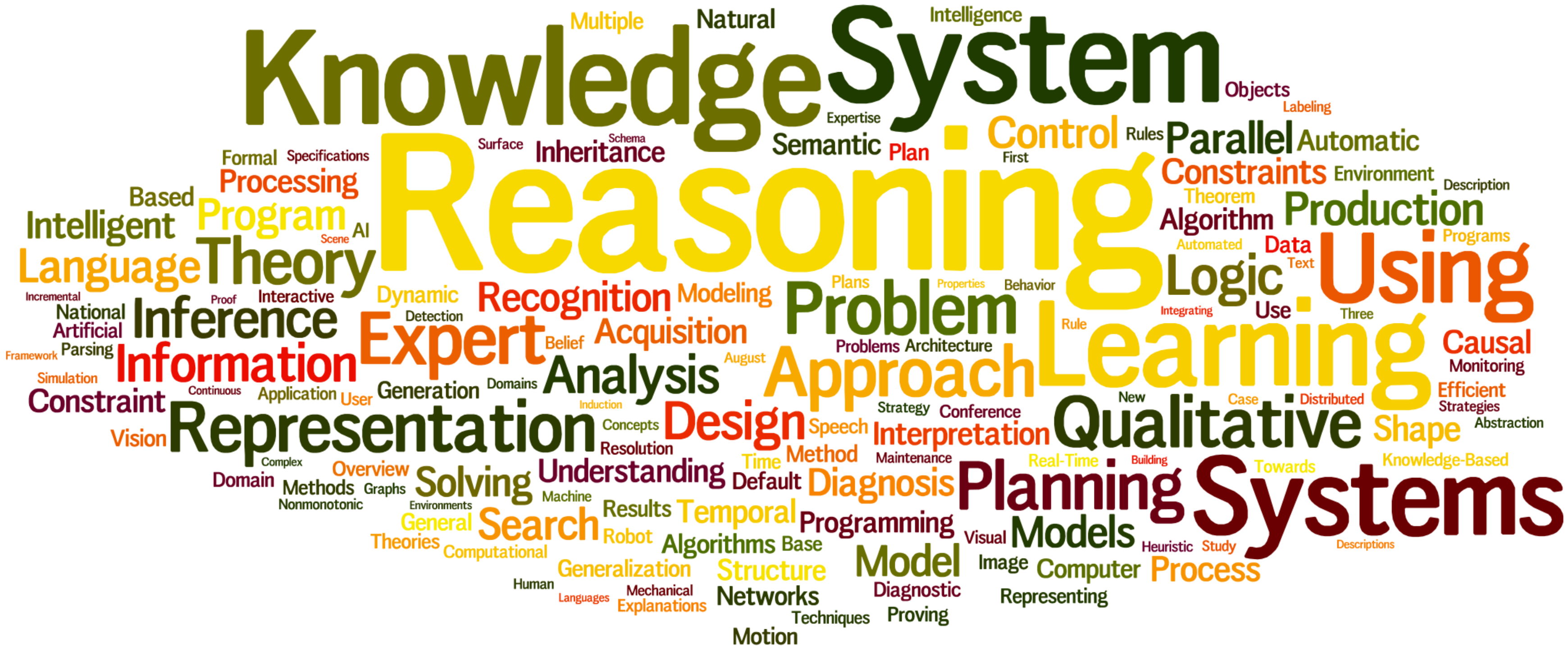
- Basis for intelligent agents framework
- Unclear if this captures the current scope of AI research

The pragmatist view

“AI is that which appears in academic conferences on AI”

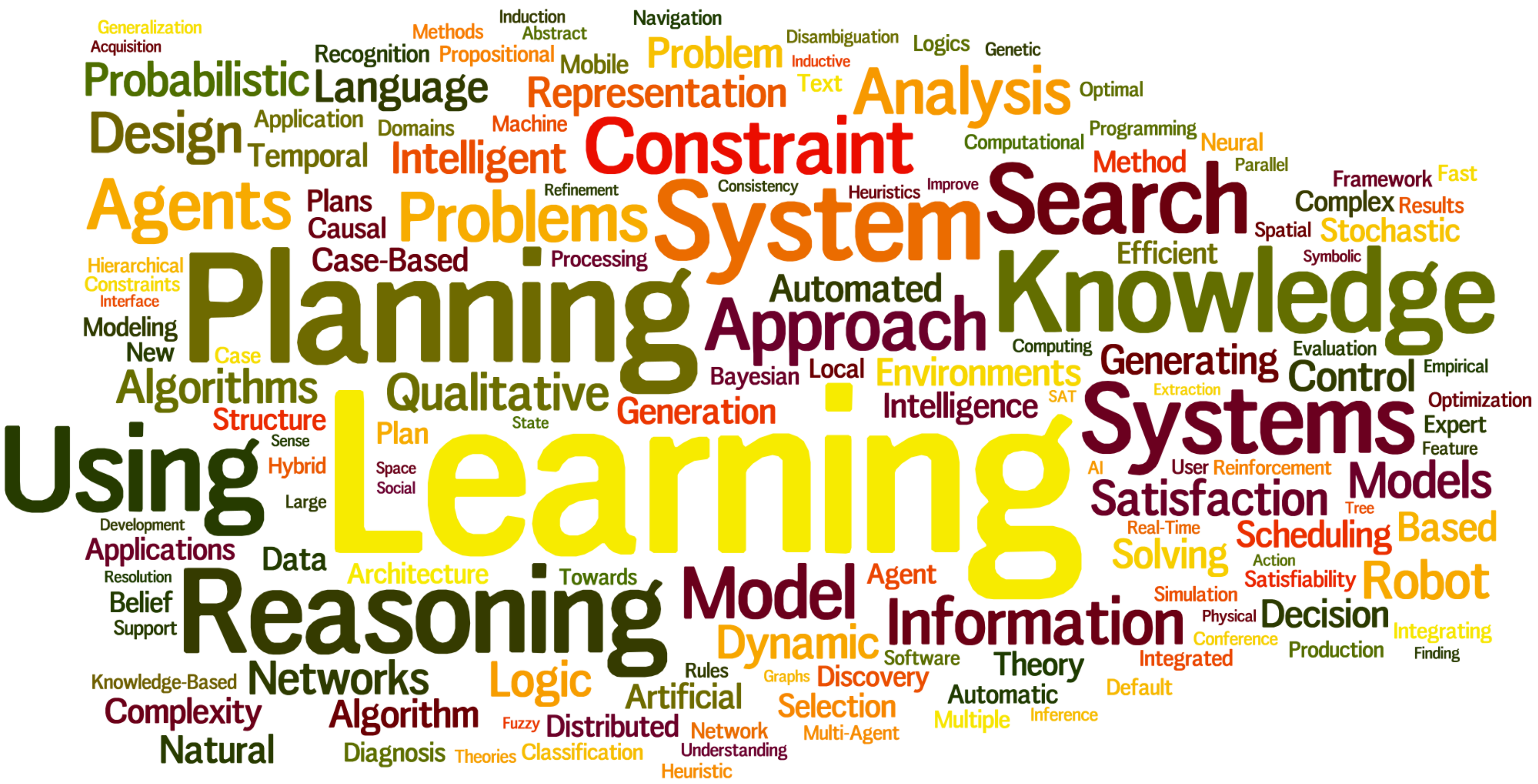
Alternate definition: “AI is that which marketing departments call AI”

Paper titles in AAAI



1980s

Paper titles in AAAI



1990s

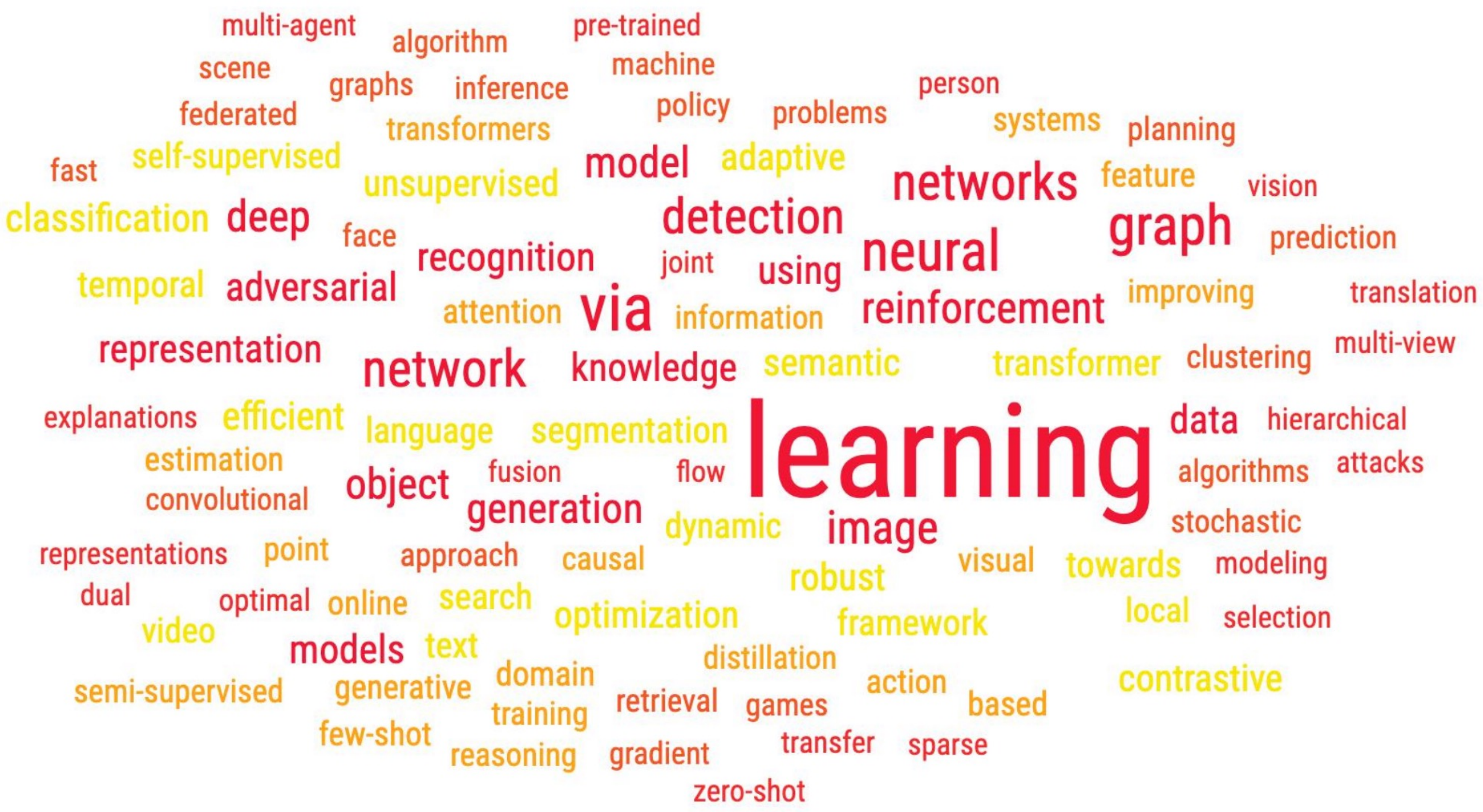
Paper titles in AAAI

2000s

Paper titles in AAAI

2010s

Paper titles in AAAI



2020s

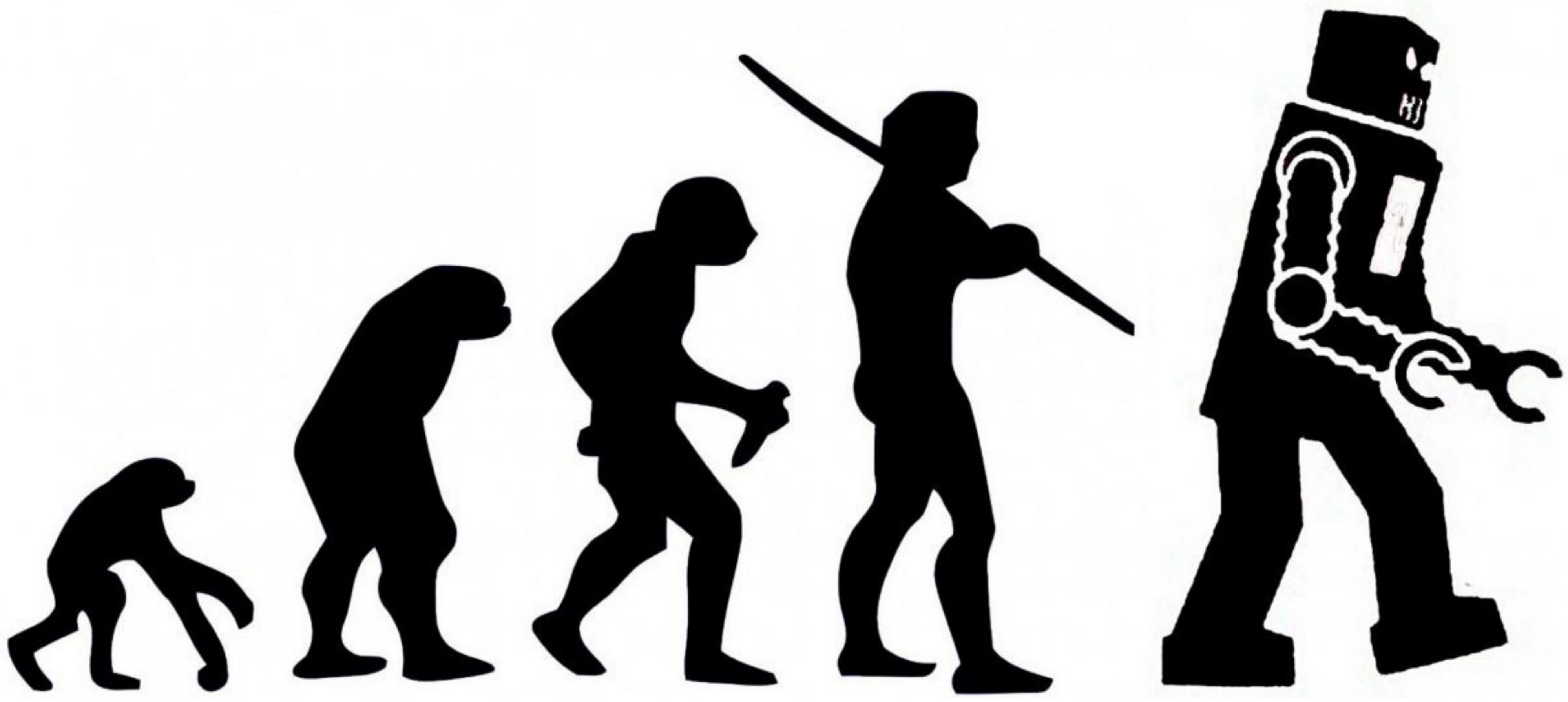
A broader (but vague) definition

Artificial intelligence is the development and study **of computer systems** to address **complex** real-world problems typically associated with some form of intelligence

Outline

- Course logistics
- What is AI?
- History of AI
- Current applications of AI
- What is an agent?

A brief history of AI



Birth of AI (1943 –1956)

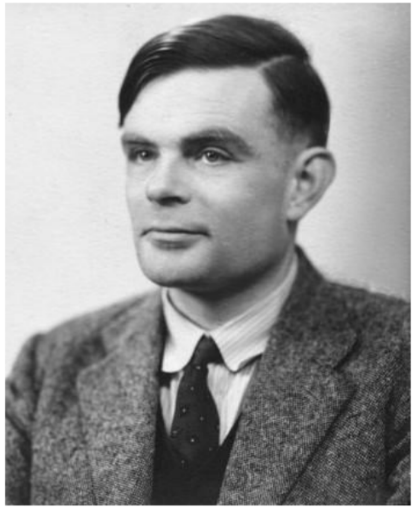
- [1943] McCullogh and Pitts: simple neural networks
  - A computational model inspired by the brain

A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY*

■ WARREN S. MCCULLOCH AND WALTER PITTS
University of Illinois, College of Medicine,
Department of Psychiatry at the Illinois Neuropsychiatric Institute,
University of Chicago, Chicago, U.S.A.

Birth of AI (1943 –1956)

- [1943] McCullogh and Pitts: simple neural networks
- [1950] Turing test



A. M. Turing (1950) Computing Machinery and Intelligence. *Mind 49:* 433-460.

COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

Birth of AI (1943 –1956)

- [1943] McCullogh and Pitts: simple neural networks
- [1950] Turing test
- [1955-56] Newell and Simon: Logic Theorist

First program deliberately engineered to perform automated reasoning; eventually goes onto being called the first artificial intelligence program

Uses search + heuristics

Birth of AI (1943 –1956)

- [1943] McCullogh and Pitts: simple neural networks
- [1950] Turing test
- [1955-56] Newell and Simon: Logic Theorist
- [1956] Dartmouth workshop (coined the term artificial intelligence)

Birth of AI (1956 Dartmouth workshop)

- 1956: workshop at Dartmouth college; 11 attendees including John McCarthy, Marvin Minsky, Claude Shannon, Allen Newell and Herbert Simon

John McCarthy

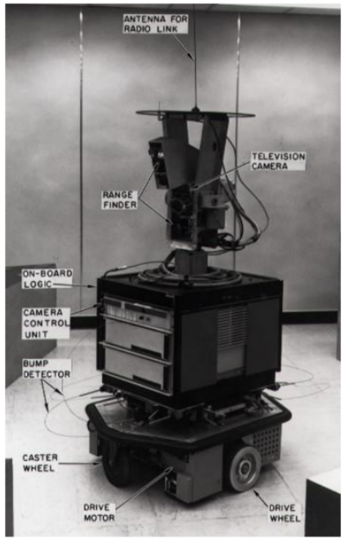
Aim for general principles: Every aspect of learning or any other feature of intelligence can be so precisely described that a machine can be made to simulate it

We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer

Early successes in AI (1950s – 60s)

[1952] Checkers: Arthur Samuel's program learned weights via self-play and played at a strong amateur level

[1958] McCarthy LISP, advice taker, time sharing



[1968-72] Shakey the robot

[1971-74] Blocksworld planning and reasoning domain

Early success in AI (1950s – 60s)

Overwhelming optimism

Machines will be capable of doing any work a man can do – Herbert Simon [1965]

Within a generation, I am convinced, few compartments of intellect will remain outside the machine's realm — the problems of creating "artificial intelligence" will be substantially solved – Marvin Minsky [1960s]

I visualize a time when we will be to robots what dogs are to humans. And I am rooting for the machines – Claude Shannon

First AI winter (Later 1970s)

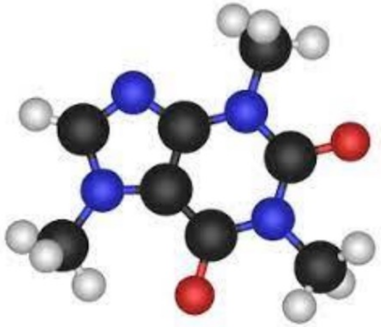
AI did not live up to the promise

- 1966 ALPAC report cut off funding for machine translation
  - *we will not suddenly or at least quickly attain machine translation*
- 1973 LightHill report
  - In no part of the field have the discoveries made so far produced the major impact that was then promised
- 1970s DARPA cut funding

What went wrong?

- Limited compute: search space grows exponentially
- Limited information about the complex world
- How to address this? The answer at the time was knowledge-based systems or expert systems that encode prior knowledge
  - Moved away from the optimism of generality…

Knowledge based systems (1970s-80s)



- [1971-74] Feigenbaum's DENRAL to infer molecular structure from mass spectrometry
- MYCIN: diagnose blood infections, recommend antibiotics
- 1981–Japan's "fifth generation" computer project, intelligence computers running Prolog
- [1982] XCON or R1 expert system to configure customer orders; deployed at DEC and saved $40 million a year

Second AI winter (late 1980s to early 1990s)

- Knowledge based systems also failed to deliver at the time
  - Required considerable manual effort to develop and maintain
  - "Knowledge acquisition bottleneck"
  - Deterministic rules could not handle uncertainty
- [1987] DARPA cuts AI funding for expert systems
- [1991] Japan's fifth generation project fails to meet goals

Splintering & changing of AI, and cross-fertilization with other fields (mid 1990s-)

- Many subfields and ideas: machine learning, computer vision, robotics, language processing, multiagent systems, ...
- Ideas from different fields
  - Bayes rule from probability
  - Cross-fertilization between search in AI and integer programming in operations research
  - Game theory from mathematics and economics
  - Stochastic gradient descent from statistics
  - Value iteration from control theory
  - Artificial neural networks from neuroscience
- AI becomes more mathematical
- Statistical rigor starts to be required in experimental results

Beyond symbolic AI

- Symbolic AI: top-down approach with a vision
- Neural AI: bottoms-up approach inspired by the human mind
- Neural AI had its own story with promise, successes, and winters

Phase one of neural AI

- [1943] McCulloch/Pitts model for computation
- [1958] Rosenblatt's perceptron algorithm for binary classification
- [1969] Perceptrons book showed that linear networks could not solve XOR <> part of the AI winter that killed neural net research

Phase two of neural AI

- [1986] Popularization of the backpropagation algorithm (originally invented by Seppo Linnainmaa in 1972) for training multi-layer networks by Rumelhardt, Hinton, Williams

- [1989] LeCun applied CNNs (pioneered previously by CMU's Alex Waibel) for handwriting recognition

- Was still hard to train very deep networks successfully---hence successes limited to simple tasks

Phase three of neural AI

- "AlexNet moment" [2012]: huge gains on a real-world image classification task by successfully training deep networks
  - Large scale data (ImageNet) + GPUs + training heuristics

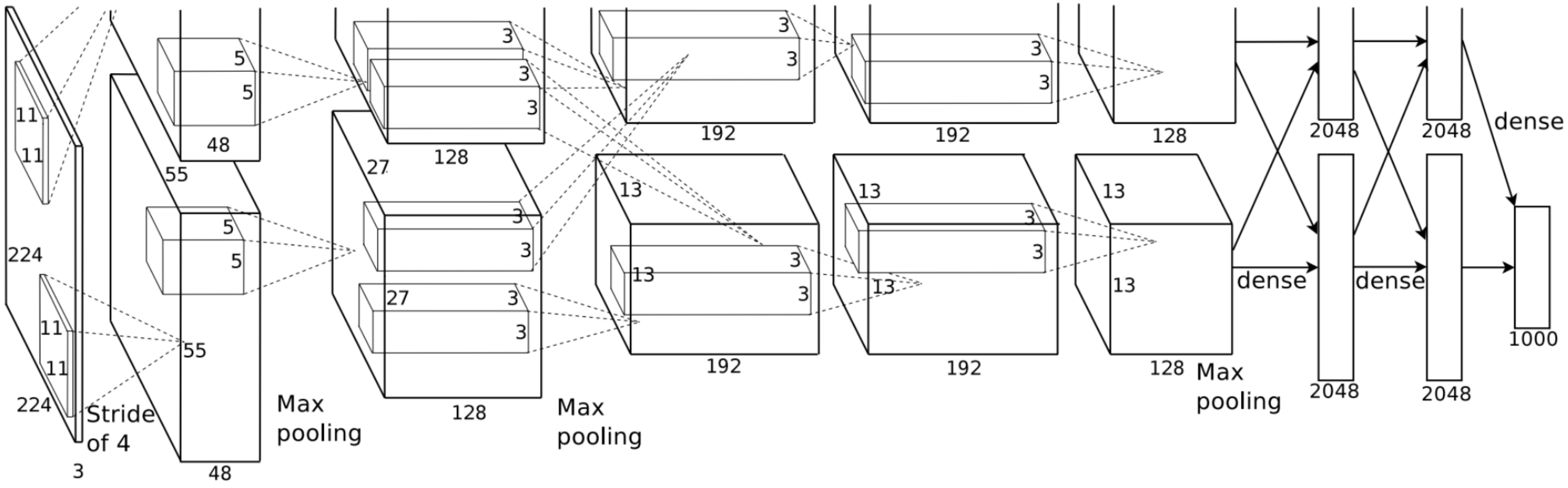


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

Phase three of neural AI

- Ever larger amounts of data and compute (both offline and real time)

- [2020] GPT-3 with 175B parameters, trained on about 45TB of text data from different datasets
- [2024] Llama-3.1 was released on July 23, 2024, with three sizes: 8B, 70B, and 405B parameters

The AI renaissance

- [1997] DeepBlue defeats Garry Kasparov

- [1995] NavLab5 automobile drives across country 98% autonomously

- [2005, 2007] Stanford and CMU win DARPA grand challenges in autonomous driving

- [2011] IBM's Watson defeats human Jeopardy opponents

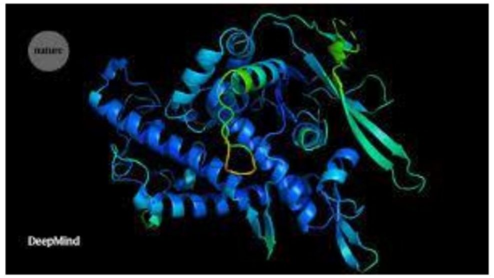
The AI renaissance

- [2016] DeepMind's AlphaGo beats top human player Lee Sodol

- [2017] CMU's Libratus defeats world's best players at two-player no-limit Texas Hold'em

- [2019] CMU's Pluribus defeats world's best players at multi-player no-limit Texas Hold'em

- [2021] DeepMind's AlphaFold offers highly accurate protein structure prediction



Outline

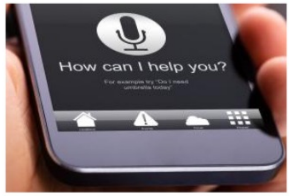
- Course logistics
- What is AI?
- History of AI
- Current applications of AI
- What is an agent?

AI in the real world

Translation

Spam detection

Voice assistant



Stock market

Facial recognition

Hiring systems

Autonomous driving

ARTIFICIAL INTELLIGENCE

AI Discovers First New Antibiotic in Over 60 Years

A landmark study discovers a new antibiotic using AI deep learning technology.

Posted December 24, 2023 | Reviewed by Abigail Fagan

KEY POINTS

- Antimicrobial resistance poses a major threat to public health.
- Researchers used AI to discover a new class of antibiotics to treat drug-resistant staph infections.
- The study used graph-based searches for chemical substructure options.

*In Prof. Sandholm's August 2021 IJCAI John McCarthy Award talk, he predicted there will be **human-only categories** in future art competitions*

An A.I.-Generated Picture Won an Art Prize. Artists Aren't Happy.

"I won, and I didn't break any rules," the artwork's creator says.

Give this article 1.5K

Jason Allen's A.I.-generated work, "Théâtre D'opéra Spatial," took first place in the digital category at the Colorado State Fair. via Jason Allen

By Kevin Roose

Sept. 2, 2022

New York Times

Large language models (LLMs) 2018-

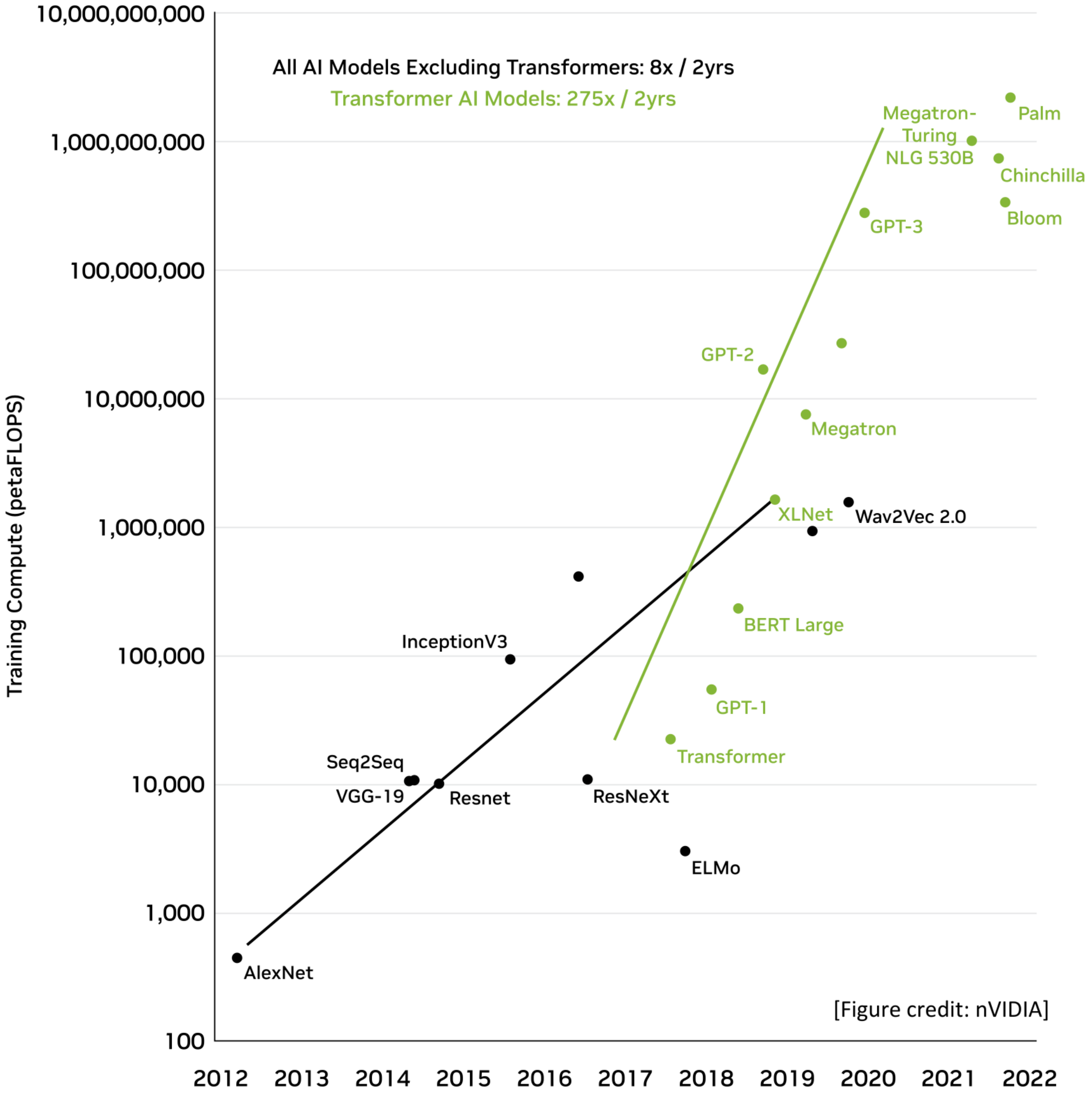
1. Generation (e.g., story writing, marketing content creation)
    - Tim Boucher wrote 97 books in 9 months using ChatGPT from OpenAI and Claude from Anthropic. He used Midjourney to generate images to match the story. Word count 2,000 - 83,000. Price $1.99 - $5.99
    - Non-AI-assisted record, by Barbara Cartland, is 191 books per year
2. Summarization (e.g., web search, legal paraphrasing, meeting notes summarization)
    - E.g., Bing co-pilot, Bard
    - In the future also question answering from specific corpora
3. Translation (e.g., between languages, text-to-code, text-to-database)
4. Classification (e.g., toxicity classification, sentiment analysis)
5. Chatbot (e.g., open-domain Q+A, virtual assistants)
    - E.g., Bing co-pilot, Bard

Downsides of LLMs

- Hallucination
  - May be impossible to fully get rid of
    - But code writing & external calls help
  - In important applications, getting sensical answers most of the time isn't good enough (cf. self-driving in 2015)
- Limited reasoning. Thus poor at
  - strategies against adversaries
    - Can't play tic-tac-toe, much less no-limit Texas hold'em poker
  - logic
  - collaborative filtering
  - …

Downsides of LLMs …

- Training cost
  - Open AI's GPT-4 took "more than $100M" [OpenAI CEO 4/17/2023] (Microsoft invested $1B+$10B in OpenAI)
  - "Really good large language models take $ Billions to train" [Amazon CEO 4/13/2023]



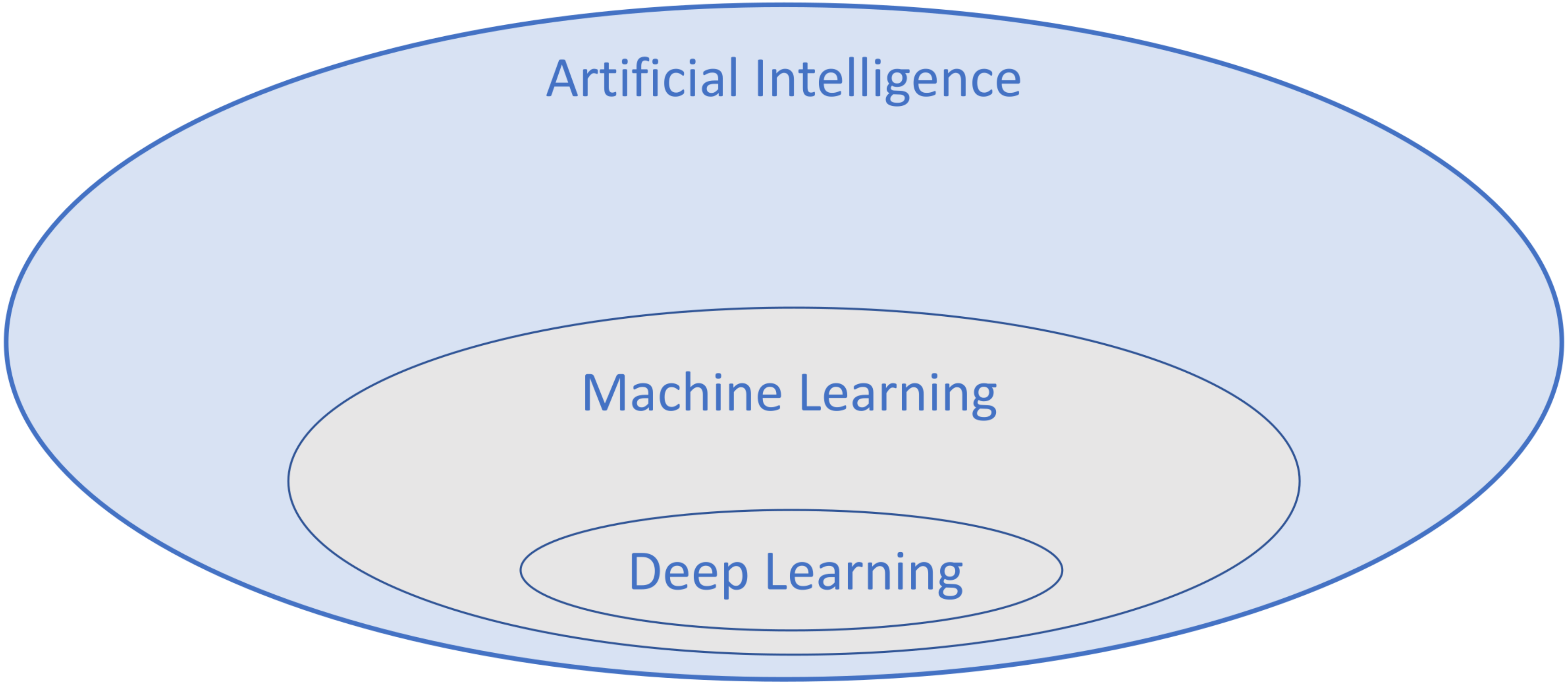
[Figure credit: nVIDIA]

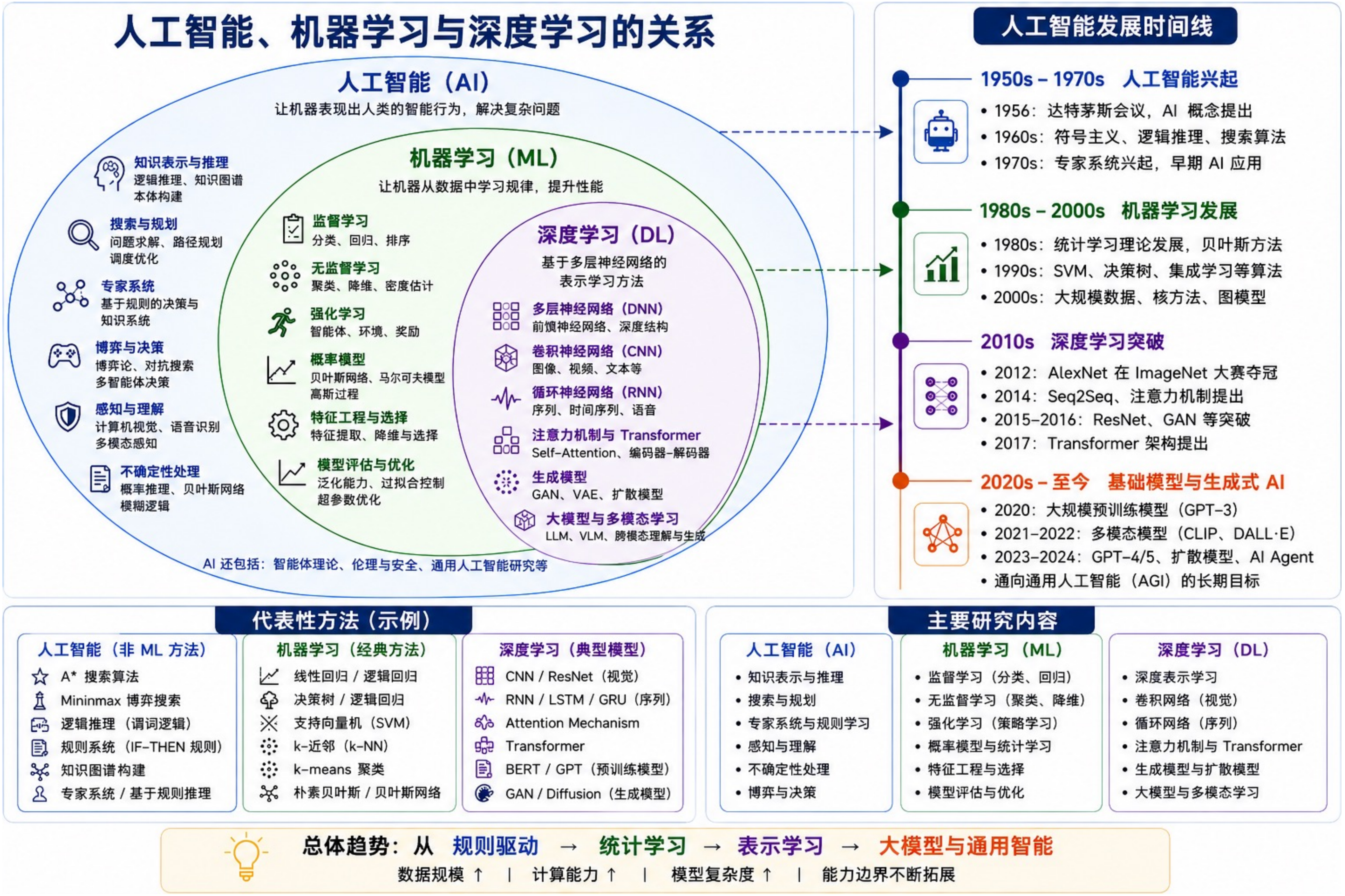
Downsides of LLMs …

- Proliferation of content
- Privacy loss & copyright issues
- Biases & inclusivity
- Explainability
- No “understanding” (?) but LLMs are used as if they understand in consulting, medicine, customer service, etc.

Regulation?

Artificial Intelligence (AI) vs Machine Learning (ML)?




人工智能、机器学习与深度学习的关系
人工智能（AI）
让机器表现出人类的智能行为，解决复杂问题
知识表示与推理
逻辑推理、知识图谱
本体构建
搜索与规划
问题求解、路径规划
调度优化
专家系统
基于规则的决策与
知识系统
博弈与决策
博弈论、对抗搜索
多智能体决策
感知与理解
计算机视觉、语音识别
多模态感知
不确定性处理
概率推理、贝叶斯网络
模糊逻辑
机器学习（ML）
让机器从数据中学习规律，提升性能
监督学习
分类、回归、排序
无监督学习
聚类、降维、密度估计
强化学习
智能体、环境、奖励
概率模型
贝叶斯网络、马尔可夫模型
高斯过程
特征工程与选择
特征提取、降维与选择
模型评估与优化
泛化能力、过拟合控制
超参数优化
深度学习（DL）
基于多层神经网络的
表示学习方法
多层神经网络（DNN）
前馈神经网络、深度结构
卷积神经网络（CNN）
图像、视频、文本等
循环神经网络（RNN）
序列、时间序列、语音
注意力机制与 Transformer
Self-Attention、编码器-解码器
生成模型
GAN、VAE、扩散模型
大模型与多模态学习
LLM、VLM、跨模态理解与生成
AI 还包括：智能体理论、伦理与安全、通用人工智能研究等
人工智能发展时间线
1950s – 1970s 人工智能兴起
• 1956：达特茅斯会议，AI 概念提出
• 1960s：符号主义、逻辑推理、搜索算法
• 1970s：专家系统兴起，早期 AI 应用
1980s – 2000s 机器学习发展
• 1980s：统计学习理论发展，贝叶斯方法
• 1990s：SVM、决策树、集成学习等算法
• 2000s：大规模数据、核方法、图模型
2010s 深度学习突破
• 2012：AlexNet 在 ImageNet 大赛夺冠
• 2014：Seq2Seq、注意力机制提出
• 2015–2016：ResNet、GAN 等突破
• 2017：Transformer 架构提出
2020s – 至今 基础模型与生成式 AI
• 2020：大规模预训练模型（GPT-3）
• 2021–2022：多模态模型（CLIP、DALL·E）
• 2023–2024：GPT-4/5、扩散模型、AI Agent
• 通向通用人工智能（AGI）的长期目标
代表性方法（示例）
人工智能（非 ML 方法）
A* 搜索算法
Mininmax 博弈搜索
逻辑推理（谓词逻辑）
规则系统（IF-THEN 规则）
知识图谱构建
专家系统 / 基于规则推理
机器学习（经典方法）
线性回归 / 逻辑回归
决策树 / 逻辑回归
支持向量机（SVM）
k-近邻（k-NN）
k-means 聚类
朴素贝叶斯 / 贝叶斯网络
深度学习（典型模型）
CNN / ResNet（视觉）
RNN / LSTM / GRU（序列）
Attention Mechanism
Transformer
BERT / GPT（预训练模型）
GAN / Diffusion（生成模型）
主要研究内容
人工智能（AI）
• 知识表示与推理
• 搜索与规划
• 专家系统与规则学习
• 感知与理解
• 不确定性处理
• 博弈与决策
机器学习（ML）
• 监督学习（分类、回归）
• 无监督学习（聚类、降维）
• 强化学习（策略学习）
• 概率模型与统计学习
• 特征工程与选择
• 模型评估与优化
深度学习（DL）
• 深度表示学习
• 卷积网络（视觉）
• 循环网络（序列）
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• 生成模型与扩散模型
• 大模型与多模态学习
总体趋势：从 规则驱动 → 统计学习 → 表示学习 → 大模型与通用智能
数据规模 ↑ | 计算能力 ↑ | 模型复杂度 ↑ | 能力边界不断拓展

下节课：机器学习基础